화학특강 (2012년 2학기)
제5장 신물질의 개발과 그 응용

중요 학습 내용

- 양자점의 메커니즘과 그 응용
- 탄소의 동소체: C60, carbon nanotubes, and graphenes
- 초임계 CO_2의 성질과 그 응용
- Ionic liquids

Origins of size effects in nanoparticles

- Origins of size effects in nanoparticles
  1. Surface effect

n: number of atoms along an edge
N: total number of atoms in the NP ($N=n^3$)

$$F = \frac{\text{number of suface atoms}}{\text{number of total atoms}} = \frac{6n^2-12n+8}{n^3} = \frac{6}{N^{1/3}}\left(1-\frac{2}{N^{1/3}}+\frac{8}{6N^{2/3}}\right) \approx \frac{6}{N^{1/3}}$$

$$\mu_{\text{nano}}(r) = \mu_{\text{bulk}}(\infty) + \frac{2\gamma \bar{V}(s)}{r}$$

  2. Quantum confinement
     - Particle in a sphere

$$E_n = \frac{n^2h^2}{8m_eR^2}$$

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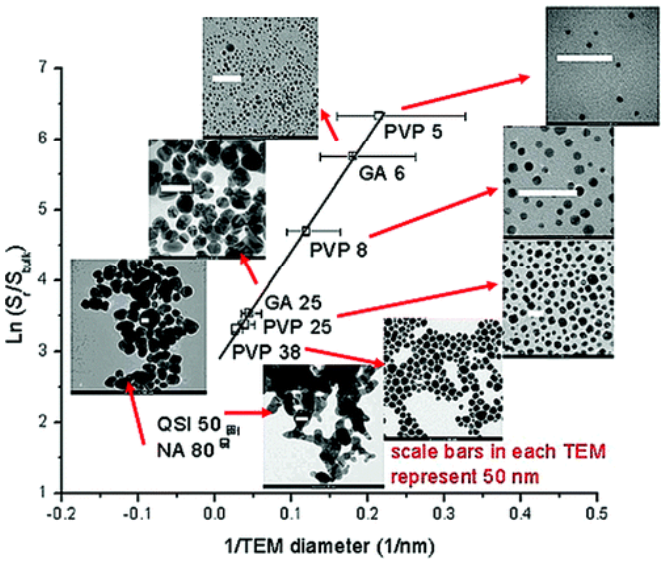
Solubility of ions and Kelvin equation

- Kelvin equation for solubility

$$x_A = x_A^* \exp\left[\frac{2\gamma \bar{V}(s)}{rRT}\right]$$

x_A^*: mole fraction of A with a flat surface
x_A: mole fraction of A with radius r

- solubility of silver nanoparticles

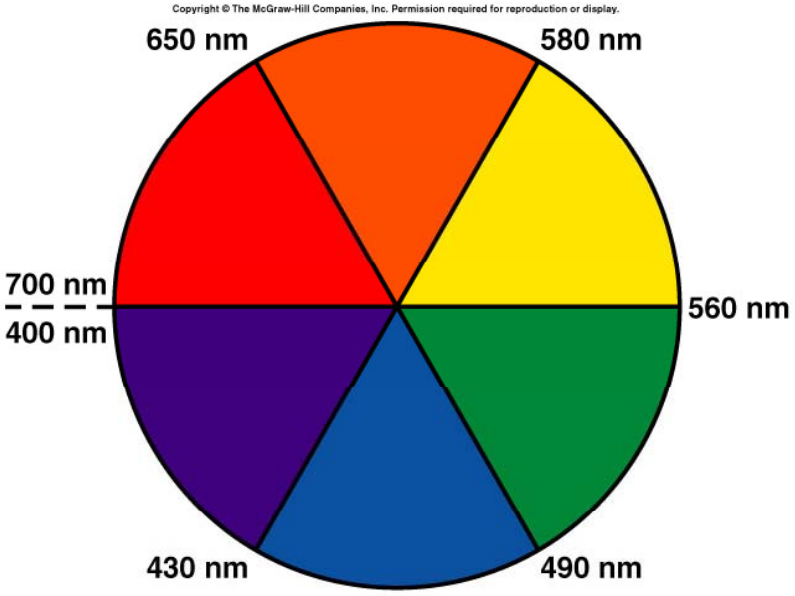


Ma et al., Environ. Sci. Technol. **46**, 752 (2012).

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Complementary Colors

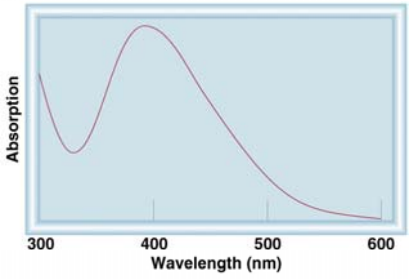
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Absorption and Fluorescence

- Absorption spectrum of bromine



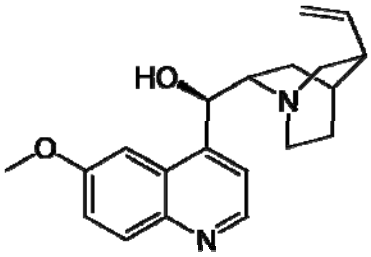


$Br_2(aq) + HCOOH(aq) \rightarrow 2Br^- + 2H^+(aq) + CO_2(g)$

Chang, 13.2

- Fluorescence of quinine



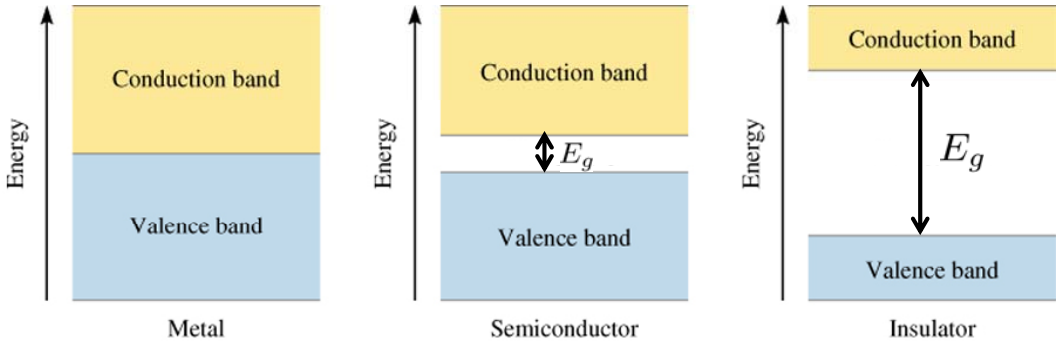


quinine

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Semiconductor

- 반도체 (semiconductor): 전도체 (conductor)와 부도체 (insulator) 사이에 존재하는 물질로서, 일정한 조건에서 도체가 되는 물질.



E_g: energy gap or band gap

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Quantum Dots (QD) – an Artifical Atom

- Quantum dot (양자점): 10 -10^5 개의 원자로 이루어진, 2 - 10 nm 지름의 반도체 나노결정.

- Energy gaps in quantum dots:

$$E_{gap}^{QD} = E_{gap}^{bulk} + \frac{h^2}{8R^2}\left(\frac{1}{m_e^*} + \frac{1}{m_h^*}\right) - \frac{1.8e^2}{4\pi\epsilon_0\epsilon R}$$

Energy gap of semiconductor in bulk state (CdSe: 1.73 eV)

- Energy of a particle in a sphere:

$$E_n = \frac{n^2h^2}{8m_eR^2}$$

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Quantum Dots (QD) – an Artifical Atom

- Exciton: bound electron-hole pair (EHP)
- Excite semiconductor → creation of EHP
  - There is an attractive potential between electron and hole
  - $m_h^* > m_e^* \Rightarrow$ hydrogenic syetem
  - Binding energy determined from Bohr Theory

$$E_n = -\frac{e^2}{8\pi^2\epsilon_0 a_0 n^2}$$

- In QDs, excitons generated inside the dot
- The excitons confined to the dot
  - Degree of confinement determined by dot size
  - Discrete energies

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Effective masses and dielectric constants in semiconductors

TABLE I. Material parameters used in the PPM calculations (Ref. 20).

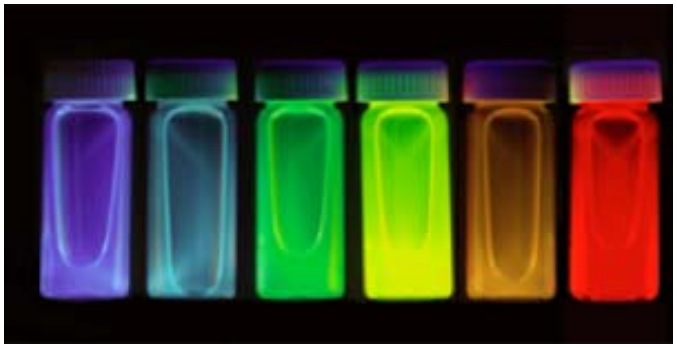
QD material	m_e^*/m_e	m_h^*/m_e	ε
CdS	0.18	0.53	5.23
CdSe	0.13	0.3	6.23
CdTe	0.11	0.35	7.1
InP	0.065	0.4	10.6
PbSe	0.07	0.06	25.0
InAs	0.028	0.33	12.3

Baskoutas & Tertis., J. Appl. Phys. **99**, 013708 (2006).

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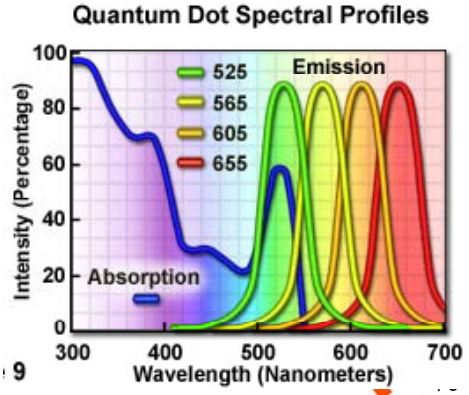
Quantum Dots (QD) – an Artifical Atom

- Fluorescence of quantum dots



Increase in quantum dot size

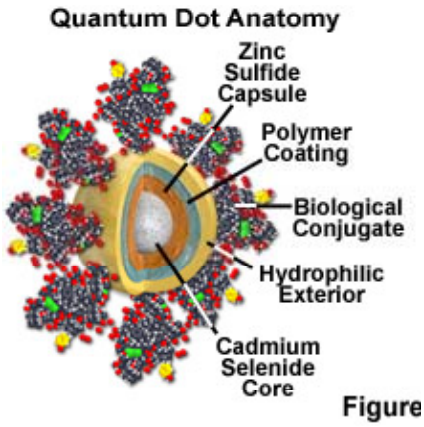
Colloidal CdSe quantum dots dispersed in hexane. Quantum confinement effects allow quantum-dot color to be tuned with particle size. (core size from 2.3 to 5.5 nm)



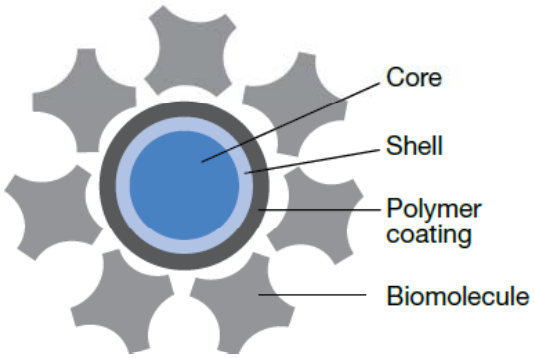
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Quantum Dots (QD) – an Artifical Atom

- Interior of CdSe quantum dots



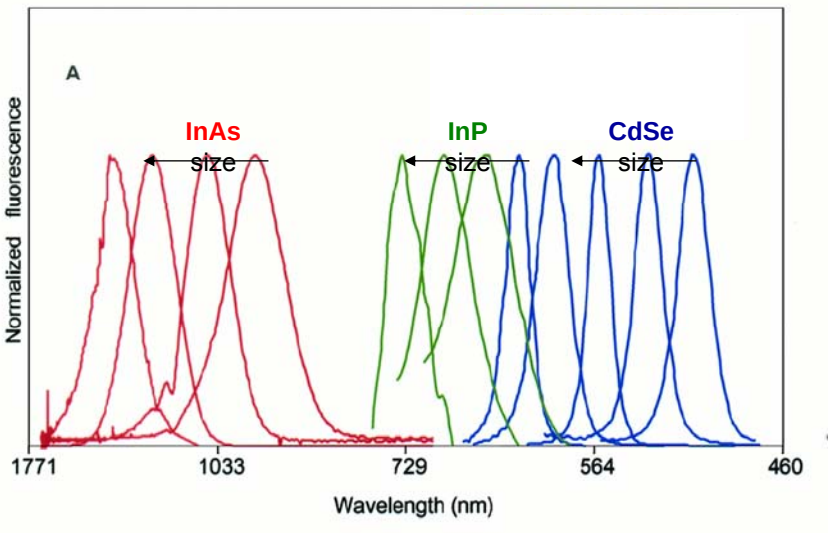
Figure

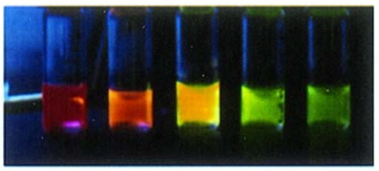


- Two disadvantages of CdSe QDs
  1. High toxicity – coating the QD's in ZnS to prevent the CdSe crystal from dissolving
  2. Insolubility in water – coating with a water-soluble hydrophilic polymer or silica

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Fluorescence Spectra of Quantum Dots





Water-soluble silica-coated Core (CdSe)-Shell (ZnS or CdS)

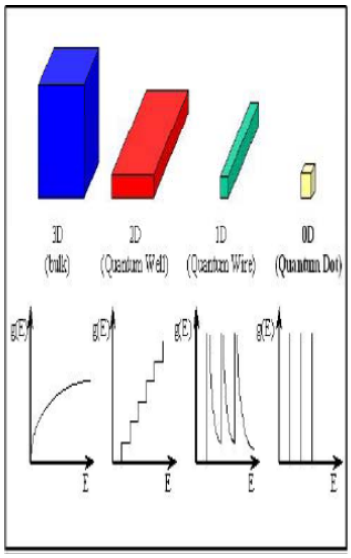
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Characteristics of Quantum Dots (QD)

- Quantum dots are small devices that contain a tiny droplet of free electrons.
- Typical dimensions are between nanometers to a few microns.
- A quantum dot can have anything from a single electron to a collection of several thousands.
- The size ,shape and number of electrons can be precisely controlled.
- Just as in an atom, the energy levels are quantized due to the confinement of electrons.
- The 3D spatial confinement is observed in the quantum dots.
- Unlike atoms however, quantum dots can be easily connected to electrodes and are therefore excellent tools to study atomic-like properties…..

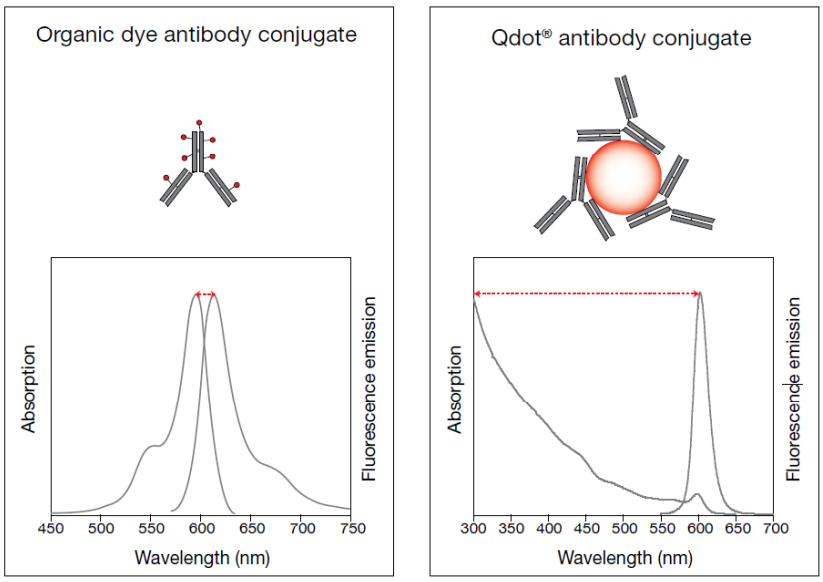
History of Quantum Dots

- In the 1970s the first low dimensional structures Quantum Wells were developed.
- 1D (quantum wires) and 0D (quantum dots) were subsequently developed.

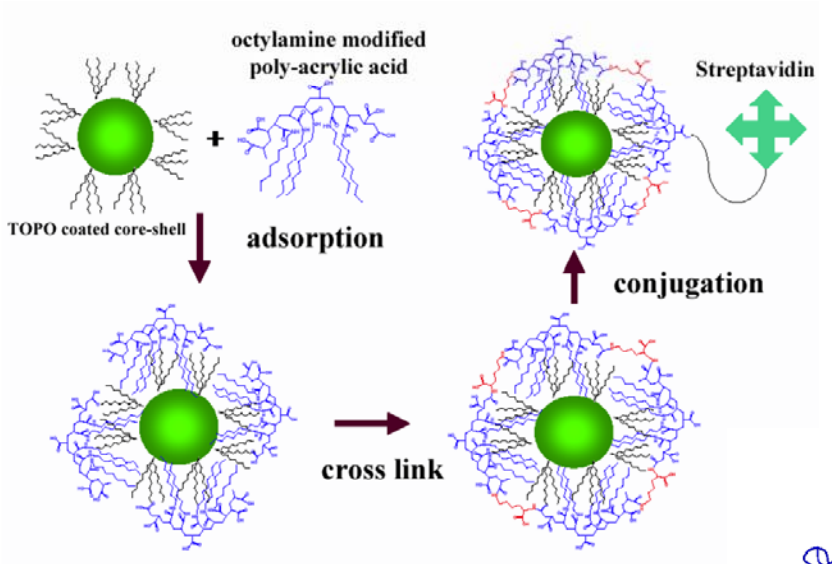


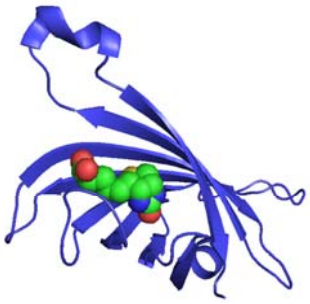
Quantum Dots and organic dyes

- Quantum dots in biology and medicine
  - Advantages of QD over organic dye molecules: organic dye molecules bleach within a minute or two, while QD's do not bleach and they are also about 10- to 20-times as bright as commonly used organic fluorophores.

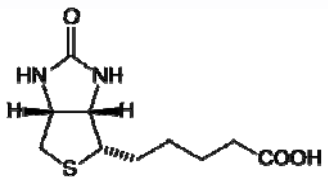


Biomolecule conjugation of quantum dots

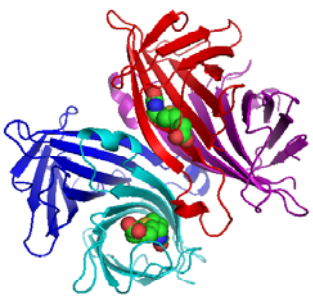




Steptavidin monomer with a bound biotin

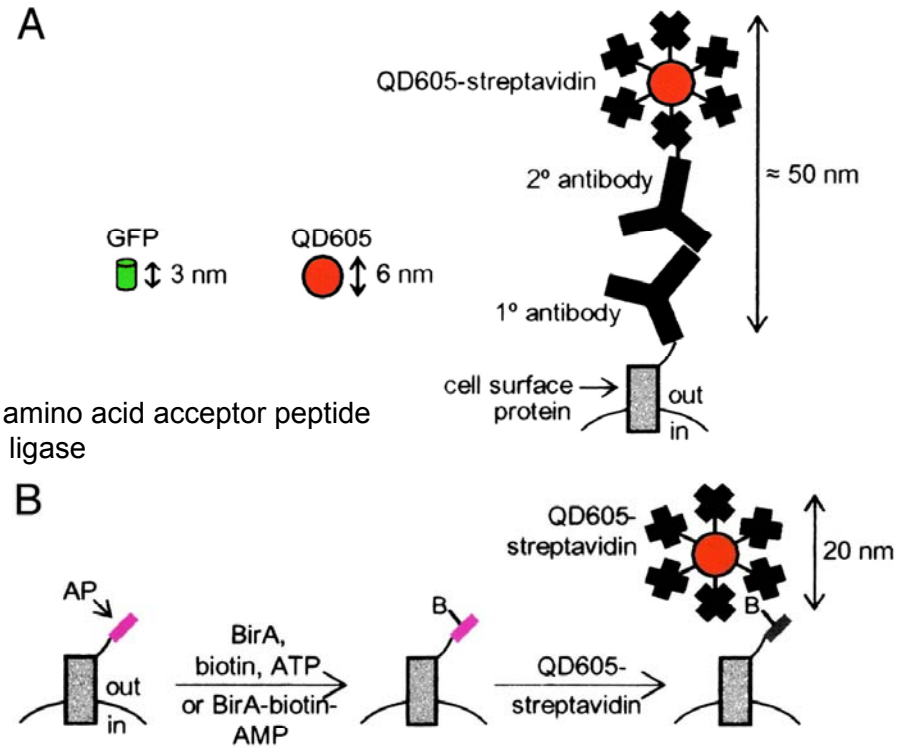


Biotin



Steptavidin tetramer with two bound biotins

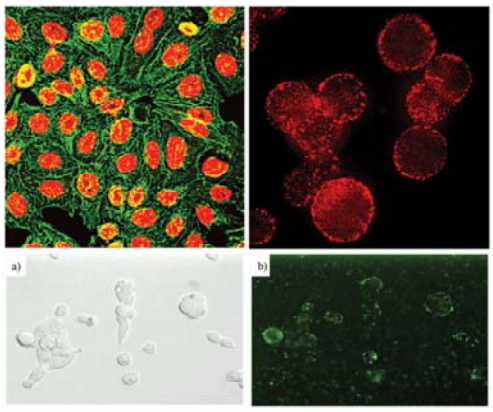
Biomolecule conjugation of quantum dots



AP: fifteen amino acid acceptor peptide
BirA: biotin ligase

Howarth et al., Proc. Natl. Acad. Sci. **102**, 7583 (2005).

Quantum Dots and Their Applications (I)



Parak et al., Nanotechnology **16**, R9 (2005)

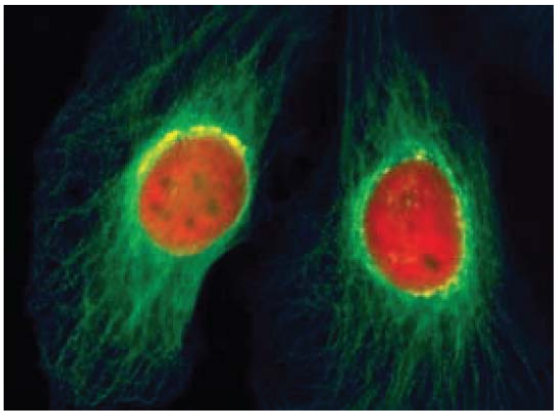
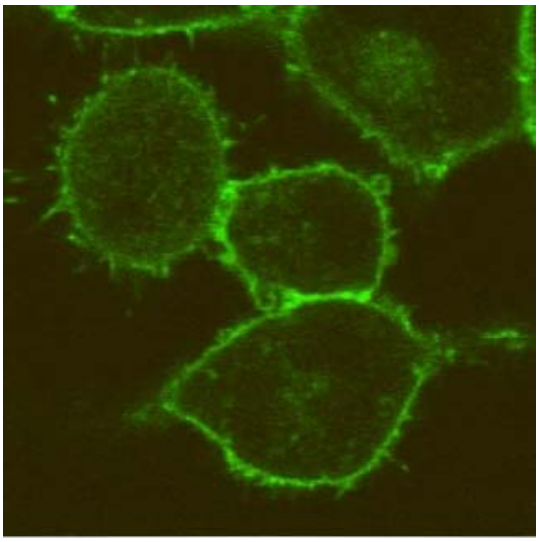


Figure 13—Multicolor immunofluorescence imaging with Qdot® secondary detection conjugates. Tubulin fibers in fixed HeLa cells were labeled with rat anti–α-tubulin antibody, biotinylated goat anti–rat IgG antibody, and Qdot® 525 streptavidin (green); Golgi bodies were labeled with rabbit anti-giantin antibody and Qdot® 585 goat F(ab')$_2$ anti–rabbit IgG antibody (yellow); and nuclei were labeled with mouse anti-nucleosome antibody and Qdot® 655 goat F(ab')$_2$ anti–mouse IgG antibody (red).

Quantum Dots and Their Applications (II)

- Quantum dots to study living cells



Lidke et al., Nature Biotechnology **22**, 198 (2004)

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Quantum Dots and Their Applications (III)

- A limitation of the fluorescent dots is that visible light frequencies would only be useful near the surface of the skin.
  - — QD's that emit near-infrared light.
  - — Near infrared penetrates tissue much more efficiently than visible light.

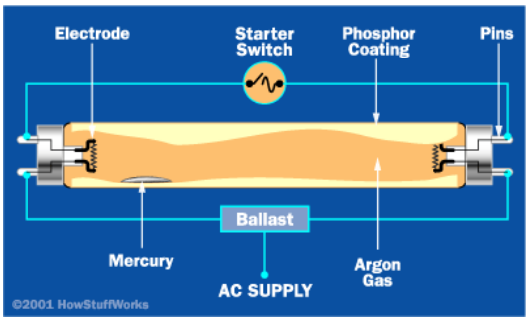


Kim et al., Nature Biotechnology **22**, 93 (2004)

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Quantum Dots and Their Applications (IV)

- Quantum dots in Electronics



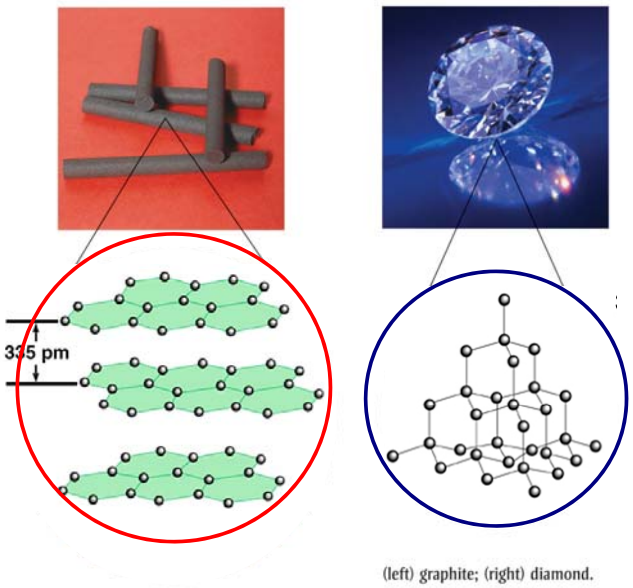
Conventional fluorescent lamp



QD fluorescent lamp - Increasing the efficiency of fluorescent lighting.

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Allotropes of Carbon – Graphite and Diamond



(left) graphite; (right) diamond.

$C_{diamond}(s) + O_2(g) \rightarrow CO_2$ $\Delta G°$=-397 kJ

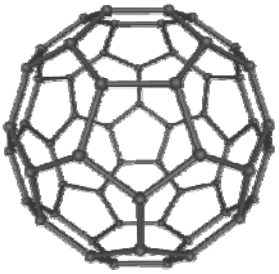
$C_{graphite}(s) + O_2(g) \rightarrow CO_2$ $\Delta G°$=-394 kJ

$C_{diamond}(s) \rightarrow C_{graphite}(s)$ $\Delta G°$ =? kJ

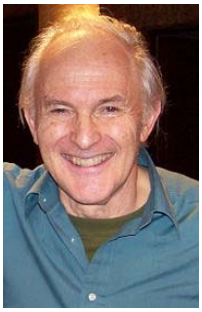
Kinetic control & Thermodynamic Control

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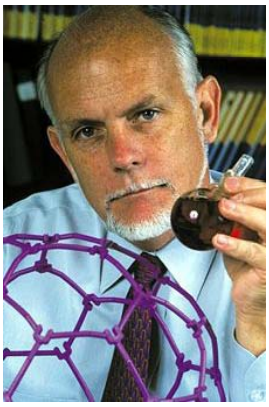
Allotropes of Carbon – Buckminsterfullerene (C_{60})



Buckminsterfullerene (6_{60})
or buckball



Harold Kroto





Geodesic dome by
R. Buckminster Fuller



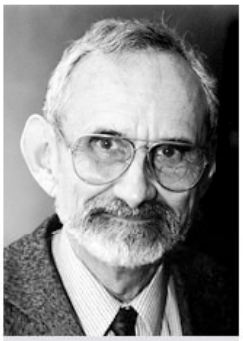
Robert Curl

Richard Smalley

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The Nobel Prize in Chemistry 1996

"for their discovery of fullerenes"



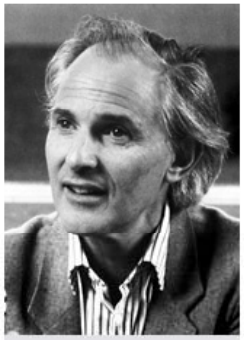
Robert F. Curl Jr.

1/3 of the prize

USA

Rice University
Houston, TX, USA

b. 1933



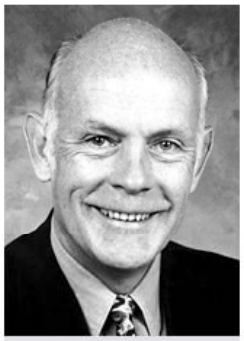
Sir Harold W. Kroto

1/3 of the prize

United Kingdom

University of Sussex
Brighton, United Kingdom

b. 1939



Richard E. Smalley

1/3 of the prize

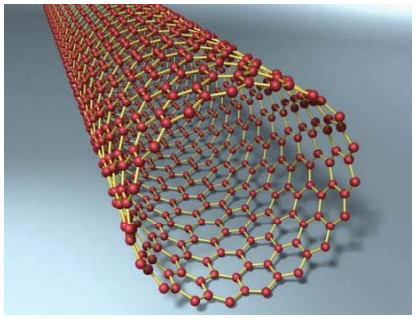
USA

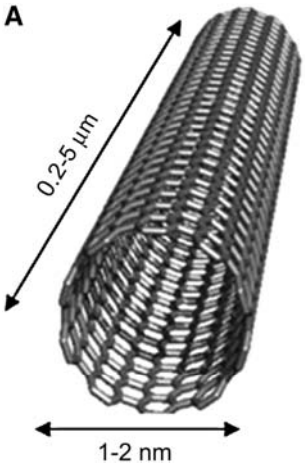
Rice University
Houston, TX, USA

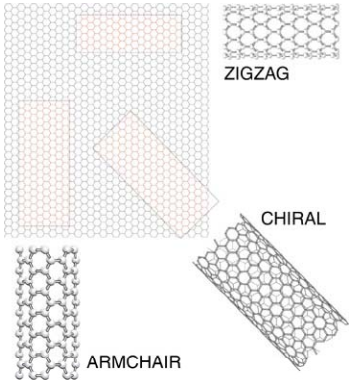
b. 1943
d. 2005

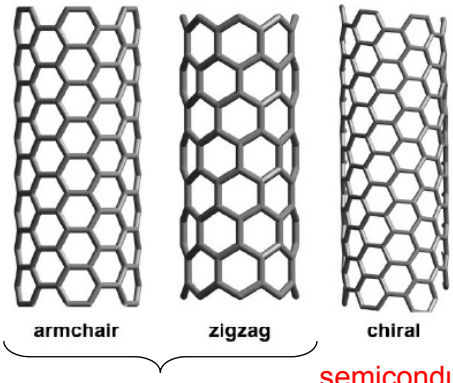
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Allotropes of Carbon – Carbon Nanotube







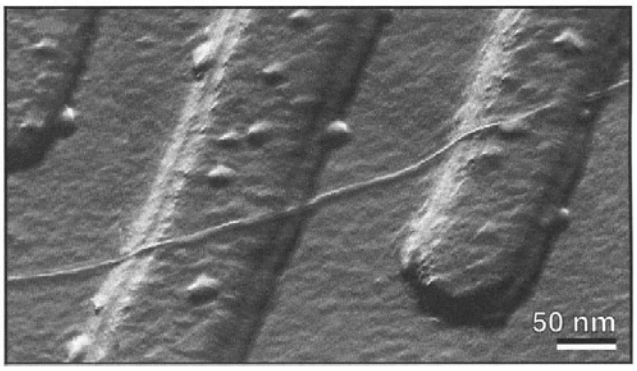


metallic, conductor

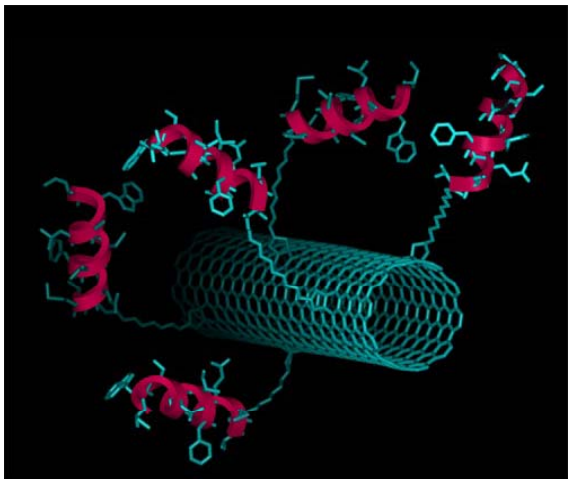
semiconductor

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Applications of carbon nanotubes



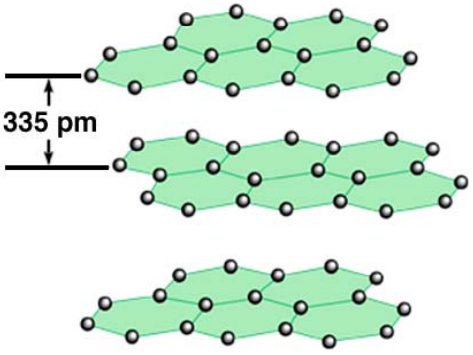
AFM tapping-mode image of a carbon nanotube on top of a Si/SiO_2 substrate (Nature 386, 474 (1997)



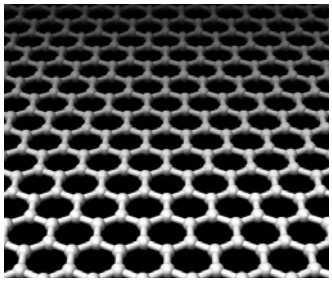
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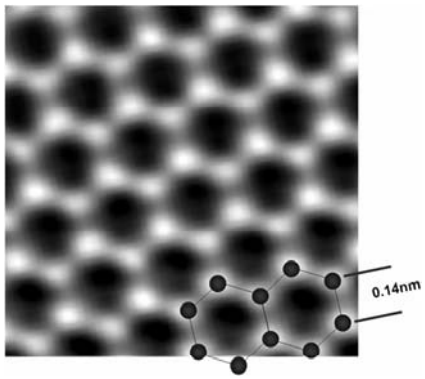
Allotropes of Carbon – Graphene

- Graphite



- Graphene (Science **306**, 666 (2004))





Transmission electron microscope (TEM) image of graphene

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The Nobel Prize in Physics 2010
Andre Geim, Konstantin Novoselov

The Nobel Prize in Physics 2010

Andre Geim

Konstantin Novoselov



Photo: Sergeom, Wikimedia Commons

Andre Geim



Photo: University of Manchester, UK

Konstantin Novoselov

The Nobel Prize in Physics 2010 was awarded jointly to Andre Geim and Konstantin Novoselov *"for groundbreaking experiments regarding the two-dimensional material graphene"*

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Tickets to the ceremony are now sold out. The event will be webcast live.

The Ig Nobel Prizes honor achievements that first make people **laugh**, and then make them **think**. The prizes are intended to celebrate the unusual, honor the imaginative — and spur people's interest in science, medicine, and technology.

"Last, but not least, there are the Ig Nobel awards. These come with little cash, but much cachet, and reward those research projects that 'first make people laugh, and then make them think'" — *Nature*

- **The winners** (the complete list, from 1991 through the present)
- **Recent news** (the Ig blog)
- **The ceremonies** (all years, including 2009 — details, video, opera librettos)
- **Some thoughts** about the Igs, about some of the winners, about what they may have done, what the Igs are, what they aren't, and what it could all mean, maybe.



The 2009 Ig Nobel Prize winners, joined by nine amused Nobel laureates, take a bow as the ceremony concludes. Photo: Richard Baguley.

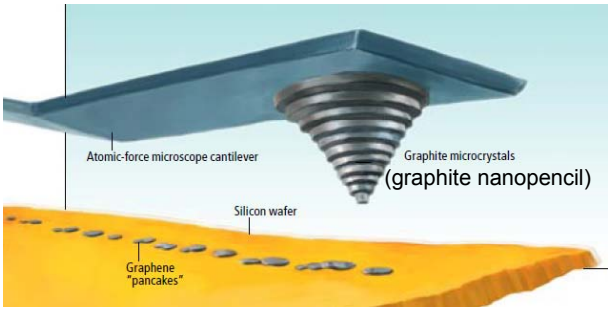
KNU

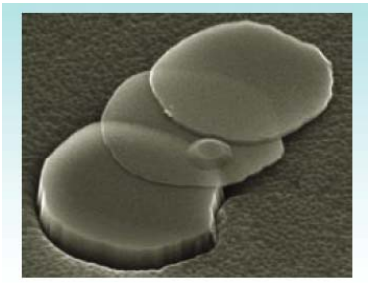
How to Make Graphenes (1)

- Using a plastic sticky tape



- Using a graphite nanopencil





Thin graphene pancakes onto a silicon wafer

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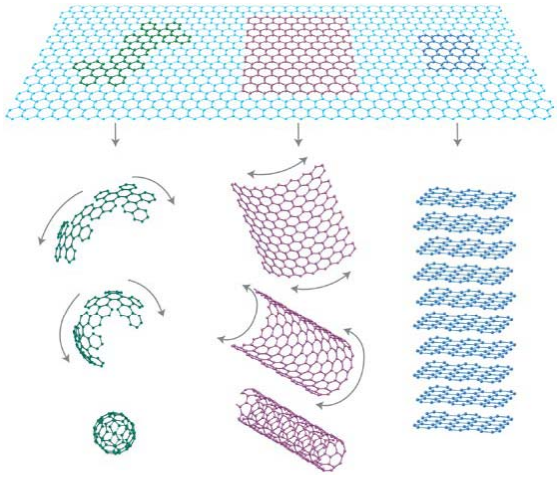
How to Make Graphenes (II)

- 기계적 박리 방법: 스카치 테잎 등을 이용하여 흑연으로부터 graphene 한 층을 벗겨내는 방법 → 좋은 품질 graphene 얻을 수 있음 → 연구 등에 이용
- Epitaxial growth: 탄화실리콘 (silicon carbide)등의 기판 위에서 수직 성장 시킴 → graphene기반의 전자 회로에 응용
- 화학적 방법: 용액상에서 흑연으로부터 산화 graphene을 만들고, 이를 환원하여 graphene을 얻음. → 대용량의 graphene 생산에 적합
  - 화학적 방법에 의해 산화 흑연이 산화 graphene으로 완전 박리
  - 용액 상에서 박리된 graphene의 분산 안전성 확보

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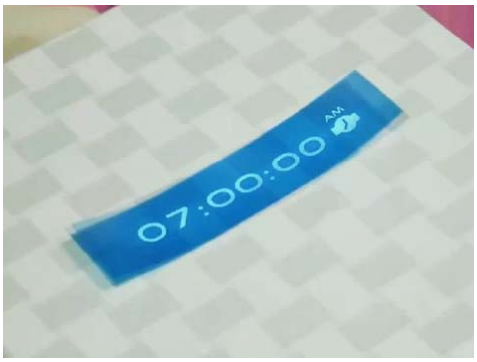
Applications of the Graphene

- Mother of most carbon allotropes



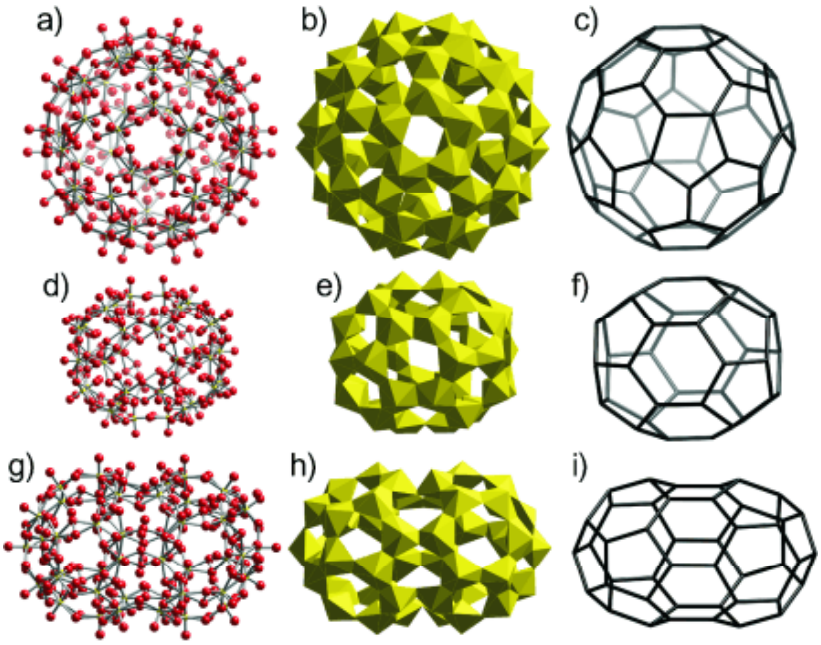
Geim, Nature Mater., **6**, 183 (2007)

- Future applications of graphene



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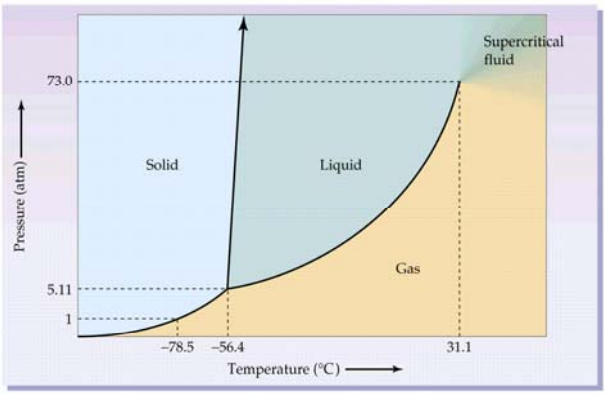
Uranyl Peroxide Fullerene



Sigmon et al., Angew. Chem. Int. Ed. **48**, 2737 (2009)

CO_2 Phase Diagram

- CO_2 Phase Diagram



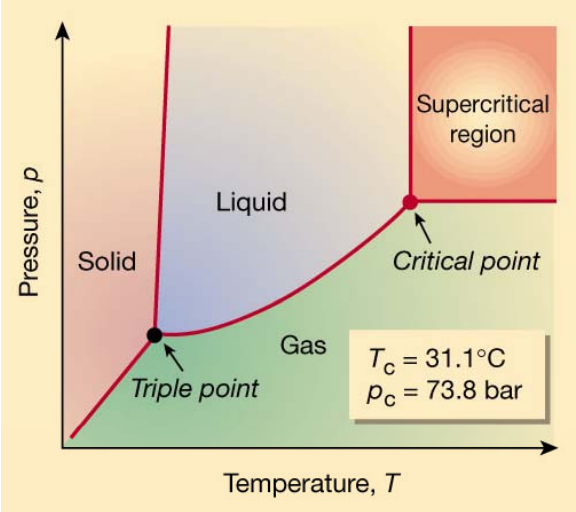


- CO_2 fire extinguisher

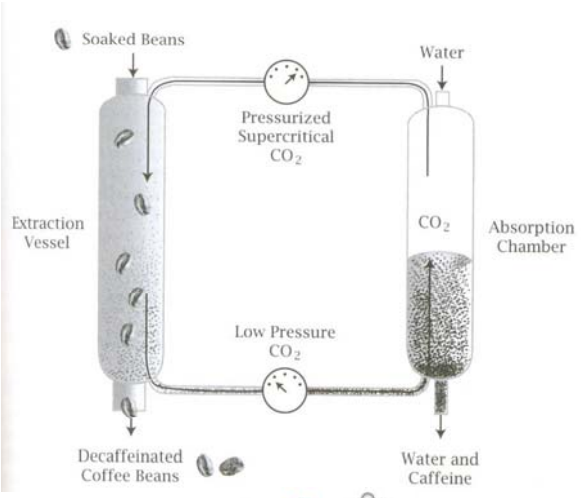


Supercritical CO_2

- Supercritical CO_2 ($scCO_2$)



- Decaffeination using $scCO_2$





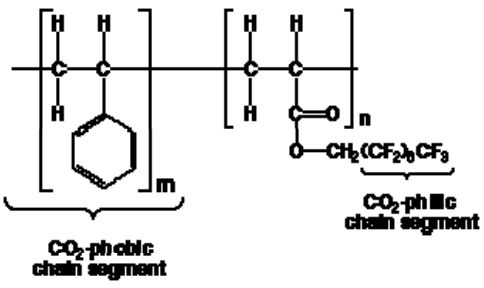
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Supercritical CO_2

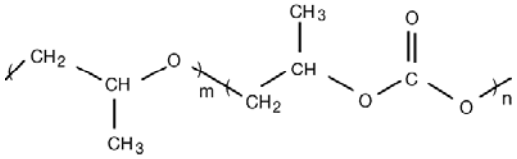
- **Advantages of $scCO_2$**
  - Ecological and economical benefits → no noxious and toxic residues once the solvent has been allowed to evaporate. (green chemistry)
  - A unique combination of properties from both liquid and gas → low viscosity and high diffusion coefficients of solutes with a density like that of a liquid.
- **Problems in $scCO_2$**
  - Poor solubility of many interesting target substances → Use of solubilizers (or stabilizer)
  - Polysiloxanes and fluorocarbons as solubilizers are expensive.
- **poly(ether_carbonate) copolymers as a solubilizer**
  - Inexpensive.
  - A favorable interaction of the carbonyl group with CO_2 may increase the enthalpy of mixing.

Atkins, Physical Chemistry 8th ed. pp119-120

- **Fluorocarbon copolymer**



- **poly(ether_carbonate) copolymers**

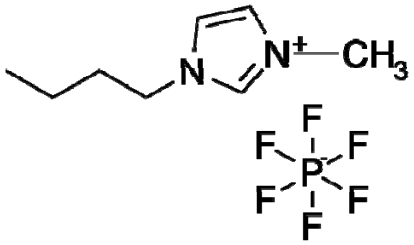


Sarbu et al., Nature 405, 165 (2000)

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Ionic Liquids

- (Room temperature) Ionic liquids: liquids that contains essentially only ions at normal temperature.
- Properties of ionic liquids
  1. Composed of small anions such as BF_4^- (tetrafluoroborate) or PF_6^- (hexafluorophosphate) and large cations such as1-hexyl-3-methylimidazolium or 1-butyl-3-methylimidazolium.



  2. Remain liquids because the bulky, asymmetrical cations do not pack together efficiently with the smaller, symmetrical anions.
  3. Are electrically conductive and have extremely low vapor pressure.

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Ionic Liquids

- Advantages of ionic liquids
  1. Their long liquid ranges, typically from -100°C ~ 200°C.
  2. The environmentally friendly nature.
  3. Can replace organic solvents.
  4. Cations that will attract potentially harmful ions such as mercury, cadmium, uranium, and americium (the latter two found in nuclear waste materials).
  5. Cations that will remove H_2S (which produces SO_2 when gas is burned) and remove CO_2 from the exhaust gases of fossil-fuel-burning power plants.

- Disadvantages of ionic liquids
  1. Their cost – very expensive.

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Solubility of nonpolar gases in perfluorocarbons

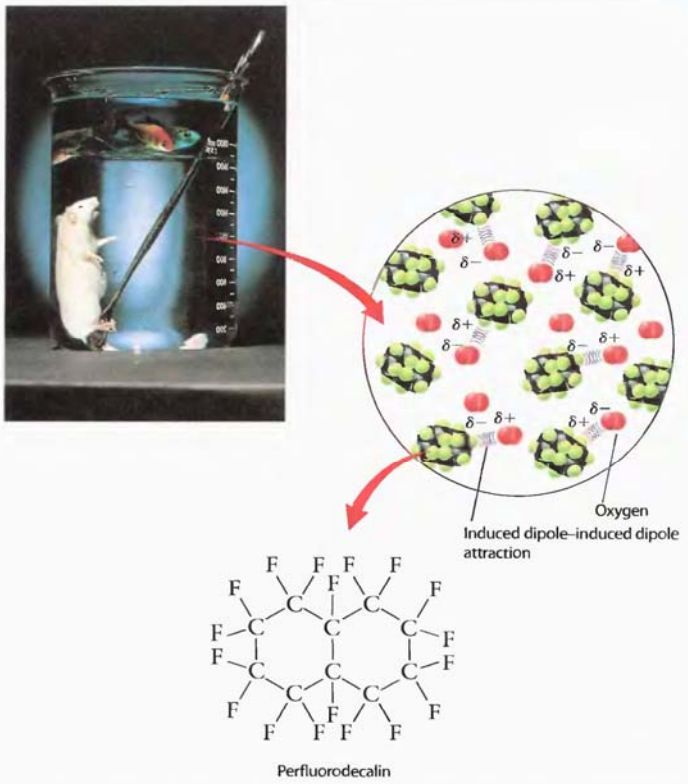


FIGURE 7.23

Just as the fish are alive and well submerged in water saturated with oxygen, this rodent is alive and well submerged in liquid perfluorodecalin saturated with oxygen. The difference is solubility. There is too little O_2 dissolved in the water for the rodent, who would drown if submerged in it. Similarly, there is too much oxygen dissolved in the perfluorodecalin for the fish, who would suffer from oxygen poisoning if they were to swim downward to say hello to the rodent.

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